

Proper subsignatures lose reset bits

Matched parameters; generic class counts with exceptions below.

observed	bits read			distinguishable classes (of 7)		
	P	Θ	V			
G only					2	P status
U _{xx} only					2	Θ status
Tr S only					3	P Θ V status
(G, U _{xx})					4	P/ Θ combinations
(U _{xx} , Tr S)					4	Θ /V combinations
(G, Tr S)					5	P/V combinations
$\Sigma = (G, U_{xx}, Tr S)$					7	all seven distinct

Filled = bit recovered; half = velocity-retained condition; open = lost.

Passive and resonant exception

At $Pe=0$ or $Mp=1$, Tr S merges $\{P,\Theta,P\Theta\}$ and (G, Tr S) merges $\{P,P\Theta\}$.
The G, U_{xx}, (G, U_{xx}) and (U_{xx}, Tr S) rows are unaffected.
The full three-coordinate decoder remains injective. Domain: $M>0, \rho>0, Pe\geq 0$.